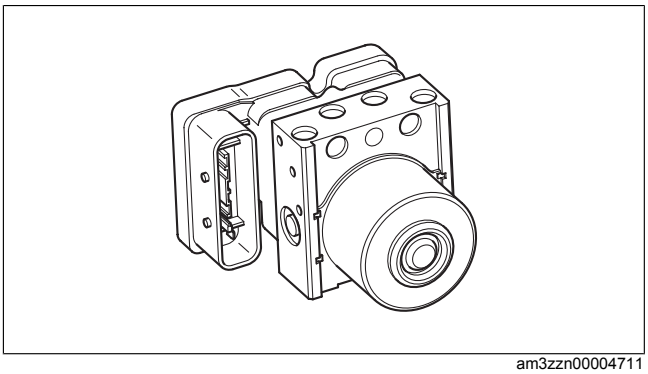
ABS HU/CM

id041300183400

Outline

- A high-reliability, reduced size and weight ABS HU/CM, integrating both the ABS HU and ABS CM, has been adopted.



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ABS HU Part Purpose/Function

- The ABS HU adjusts the fluid pressure to the caliper pistons by controlling (on/off) each solenoid valve and pump motor according to signals from the ABS CM of the ABS HU/CM.

ABS HU Part Construction

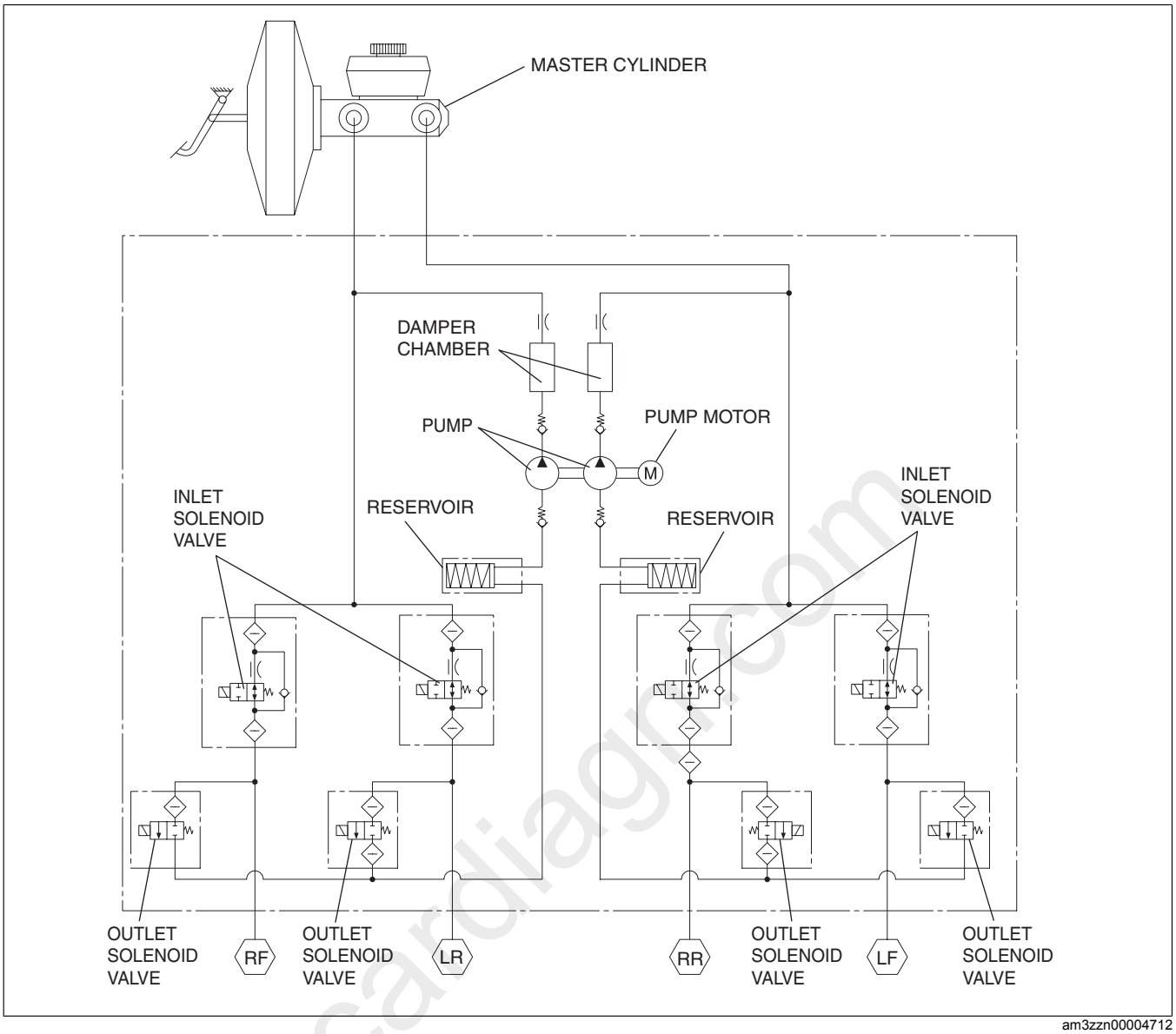
- The ABS HU mainly consists of the inlet/outlet solenoid valves, and pump motor (pump).

Function of main component parts

Part name	Function
Inlet solenoid valve	• Adjusts the fluid pressure in each brake system according to ABS CM signals.
Outlet solenoid valve	• Adjusts the fluid pressure in each brake system according to ABS CM signals.
Reservoir	• Temporarily holds the brake fluid from the caliper piston to ensure smooth pressure reduction.
Pump	• Returns brake fluid held in the reservoir back to the master cylinder.
Pump motor	• Operates the pump according to ABS CM signals.

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Hydraulic circuit diagram



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ABS HU Part Operation

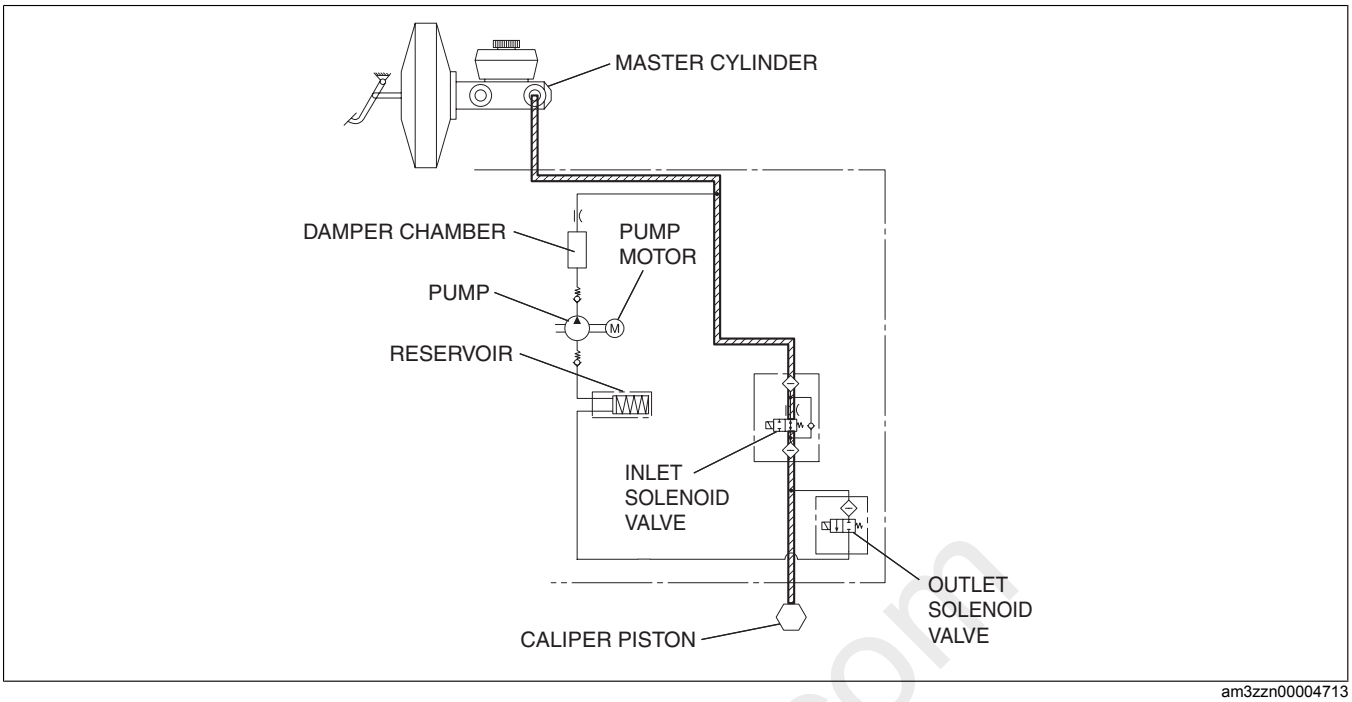
During normal braking

- During normal braking, the solenoid valves are not energized and they are all off. When the brake pedal is depressed, brake fluid pressure is transmitted from the master cylinder, through the inlet solenoid valves, and then to the caliper piston. (The figure shows control for only one, front wheel.)

Solenoid valve operation table

Inlet solenoid valve				Outlet solenoid valve				Pump motor, pump
LF	RF	LR	RR	LF	RF	LR	RR	
OFF (open)				OFF (closed)				OFF

Hydraulic circuit diagram



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During ABS and EBD control

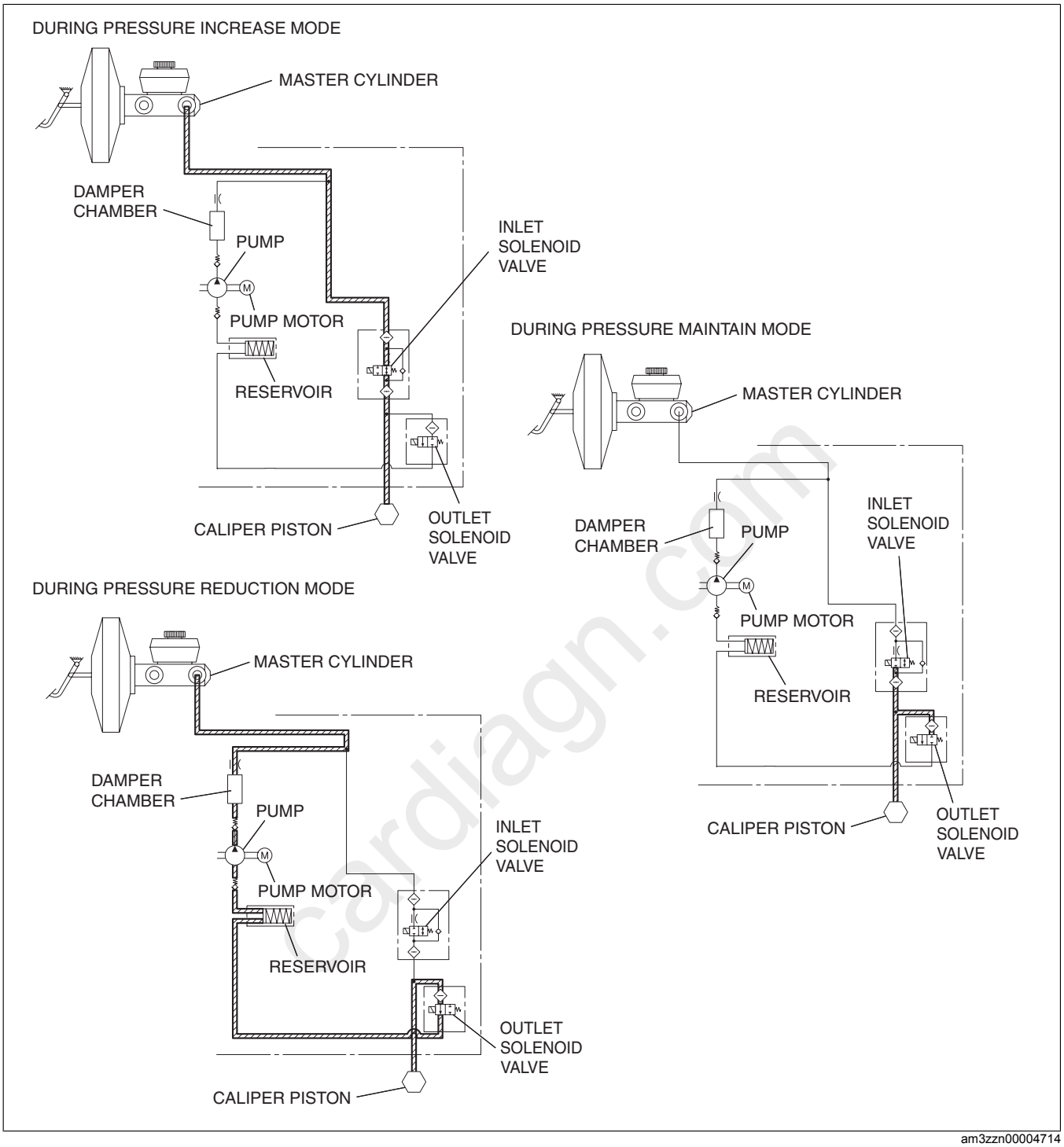
- When wheel lock-up is about to occur, the inlet and outlet solenoid valves are energized and controlled in three pressure modes (increase, maintain, or reduction), thereby adjusting brake fluid pressure. Brake fluid during pressure reduction is temporarily held in the reservoir after which the pump motor operates the pump to return the fluid to the master cylinder. (The figure shows the control for only one, front wheel)

Solenoid valve operation table

	Inlet solenoid valve				Outlet solenoid valve				Pump motor, pump
	LF	RF	LR	RR	LF	RF	LR	RR	
During pressure increase mode	OFF (open)				OFF (closed)				OFF
During pressure maintain mode	ON (closed)				OFF (closed)				OFF
During pressure reduction mode	ON (closed)				ON (open)				ON

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Hydraulic circuit diagram



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ABS CM Part Function

- The ABS CM detects the vehicle wheel speeds based on the signals from the four ABS wheel-speed sensors.
- The CM calculates the rotation condition of each wheel from the relation between the detected vehicle wheel speed and the estimated (based on the detected speed) vehicle speed from there on. It then controls brake fluid pressure to each wheel to prevent lock-up.

Function table

Function name	Contents
ABS control function	• Controls brake fluid pressure when braking to maintain directional stability, ensure steerability, and reduce stopping distance.
EBD control function	• Constantly controls proper distribution of brake fluid pressure to the front and rear wheels according to vehicle load, road surface, and vehicle speed conditions to prevent early lock-up of the rear wheels.

Function name	Contents
CAN communication function	• Outputs the vehicle wheel-speed signal and ABS system warning control data via CAN lines.
On-board diagnostic system	• Main components of the ABS control system have a self-diagnosis function. In case a malfunction occurs, warning lights illuminate to alert the driver, and at the same time, a DTC is stored in the ABS HU/CM. • When a malfunction is determined as a result of the on-board diagnostic test, system control is disabled or limited to prevent any dangerous situation while driving.
Automatic configuration function	• When the ignition is switched to ON or the engine is started after the ABS HU/CM or ABS CM has been replaced, the ABS CM reads data from the instrument cluster via CAN communication to perform automatic configuration.

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